5 AUTOMATED MACHINE LEARNING PIPELINE FOR HEAL

This chapter presents the proposed pipeline for HEAL in Section 5.1, and each of its steps. After that, we present in more detail the design of the HEAL AutoML for each of the proposed steps: Domain Understanding and Data Collection (Subsection 5.1.1, page 67), Pre-processing (Subsection 5.1.2, page 73), Data Mining (Subsection 5.1.3, page 77) and Post-processing (Subsection 5.1.4, page 79).

5.1 HEAL Pipeline

The developed HEAL system is characterized by four main steps, as presented in the Background (Chapter 2, page 19) and Methodology (Chapter 4, page 59), as follows: 1 - Domain Understanding and Data Collection ; 2 - Pre-processing; 3 - Data Mining and 4 - Post-processing. The solutions developed for each of these steps are described in the following sections. Figure 12 illustrates our proposed pipeline, highlighting the application of key techniques including ChatGPT-4o (steps represented in green and with the GPT icon), API processing, web scraping and data analysis techniques at each step of AutoML.

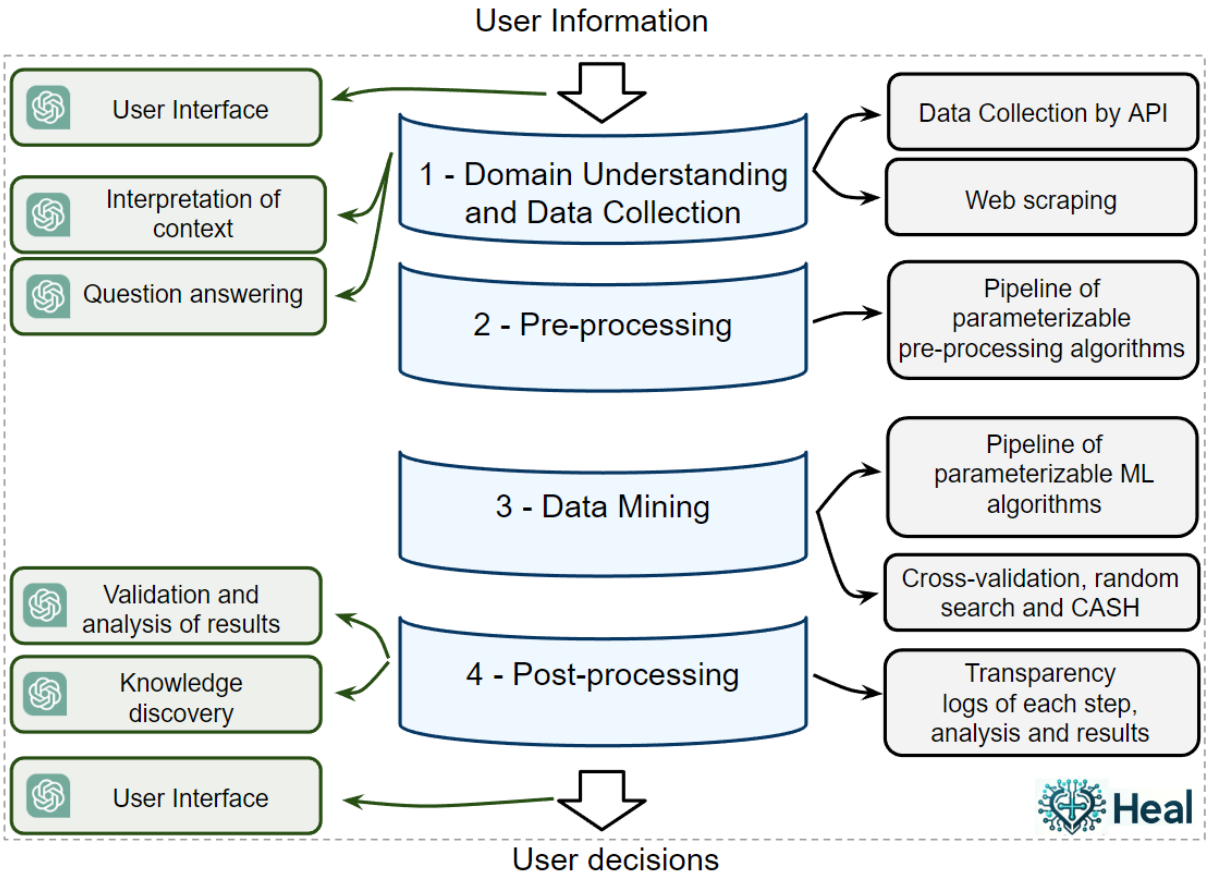
LLMs can be used to simplify and facilitate interaction between users and AutoML systems, making them more accessible tools for expert and non-expert users. Thus, this user interface improvement application was applied both in the initial step of information entered by the user, and in the final step of presenting the interpretations of knowledge generated by AutoML to the user.

Step 1: in the first step of the AutoML (Domain understanding and Data Collection) process, data analysis techniques were applied to collect data from the WHO Data API. In addition, we applied web scraping techniques on the WHO website to enrich our data related to WHO attributes.

To perform domain understanding, we use LLM techniques to interpret the context (from the most relevant indicators and information present in the dataset). Based on this interpretation, we can provide functionality to help the user understand the domain, which can answer questions the user asks related to the context.

Step 2: in the second step (pre-processing), we developed a configurable pipeline with some of the main pre-processing steps in the context of Machine Learning, for example: missing data analysis, outliers, discretization, balancing and feature selection. Thus,

Figure 12 – Steps of our pipeline (HEAL)



Source: Elaborated by the Author.

the user can configure these pre-processing steps and the pipeline executes according to this configuration. It is important to note that it is possible to add new steps in future versions of HEAL.

Step 3: in the third step of AutoML (Data Mining), we also developed a parameterizable pipeline with the Artificial Neural Networks (ANN), Random Forest (RF), Decision Tree (DT), Support Vector Machine (SVM) and K-nearest Neighbors (KNN) algorithms. All these algorithms were implemented for the context of multiclass classification problems.

For hyperparameter selection, we implement the Combined Algorithms Selection and Hyperparameter Optimization (CASH) along with cross-validation and random search. This way, the user can choose which hyperparameters will be searched and our pipeline will execute this, searching for the best combination from the defined universe. Finally, it is possible to add new algorithms in future versions of HEAL.

Step 4: LLMs techniques were also used as a tool to help validate the results found in step 4 of the AutoML process (post-processing). Thus, these algorithms received

the results, interpreted them and checked whether they are within the context defined by the user. Finally, LLMs also helped make the AutoML process more transparent and understandable for users by explaining configuration steps and decisions.

The following subsections describe in detail each of the steps performed in HEAL.

5.1.1 Step 1 - Domain Understanding and Data Collection

The first step of the HEAL pipeline is characterized by domain understanding and data collection. Therefore, as the first step of developing the system proposed in this work, we defined a solution to collect health data from WHO in the panel data structure and, consequently, build our database that will be used in the following steps. In addition to the database created by this data collection step, the user can import a dataset into our system, as long as it follows the panel data standard.

Therefore, the first step consisted of creating a database with various indicators relating to health area. Thus, as the first version of HEAL, we designed a relational database using the PostgreSQL Database Management System (DBMS).

To explore health data, the objective of the data collection step is to select and extract data from the World Health Organization (WHO). The Global Health Observatory (GHO) data repository of the World Health Organization (WHO)* is the source for the data used in this work. We used the Application Programming Interface (API) GHO Open Data Protocol.

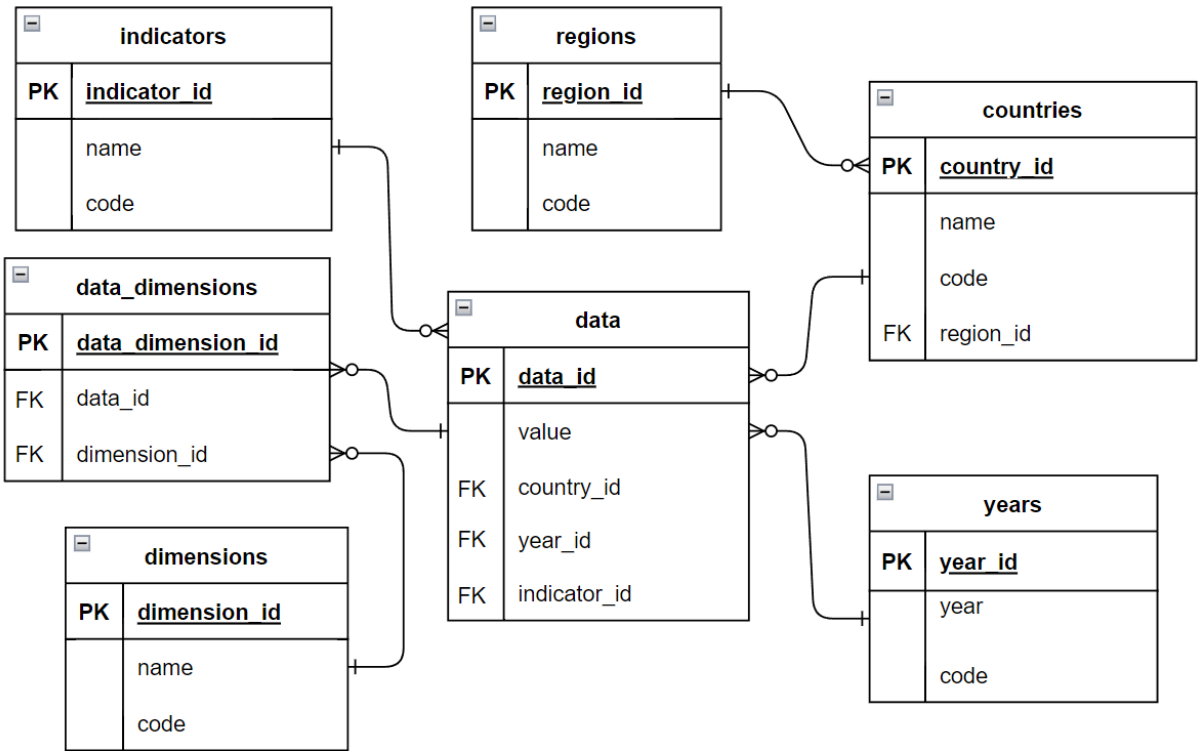
It is important to emphasize that the WHO database serves as the starting point for our research. However, HEAL step 1 is open for the user to submit new datasets in the panel data structure. This way, the user can upload a .csv file and the system will extract and insert the data from the file into the HEAL database, consequently increasing the opportunities for knowledge discovery.

To collect data, we created a HEAL step capable of searching automated (through an API) and processing automated the selected information in the WHO data repository. Then, the system inserted it in an organized way into the database, which has the structure of the illustrated relational model in Figure 13. It is possible to observe that the panel data structure is represented by the spatial dimension (regions or countries) and temporal dimension (years). Other dimensions can be expanded, for example: sex and age groups.

The purpose of the tables and relationships created is to organize the data imported from WHO into an ideal format for forecasting annual data (temporal dimension). Another feature is to scale the amount of data stored as new data is inserted without

*https://www.who.int/en/. Accessed March 15, 2024

Figure 13 – Relational Model SQL



Source: Elaborated by the Author.

losing the relationships necessary to perform the queries.

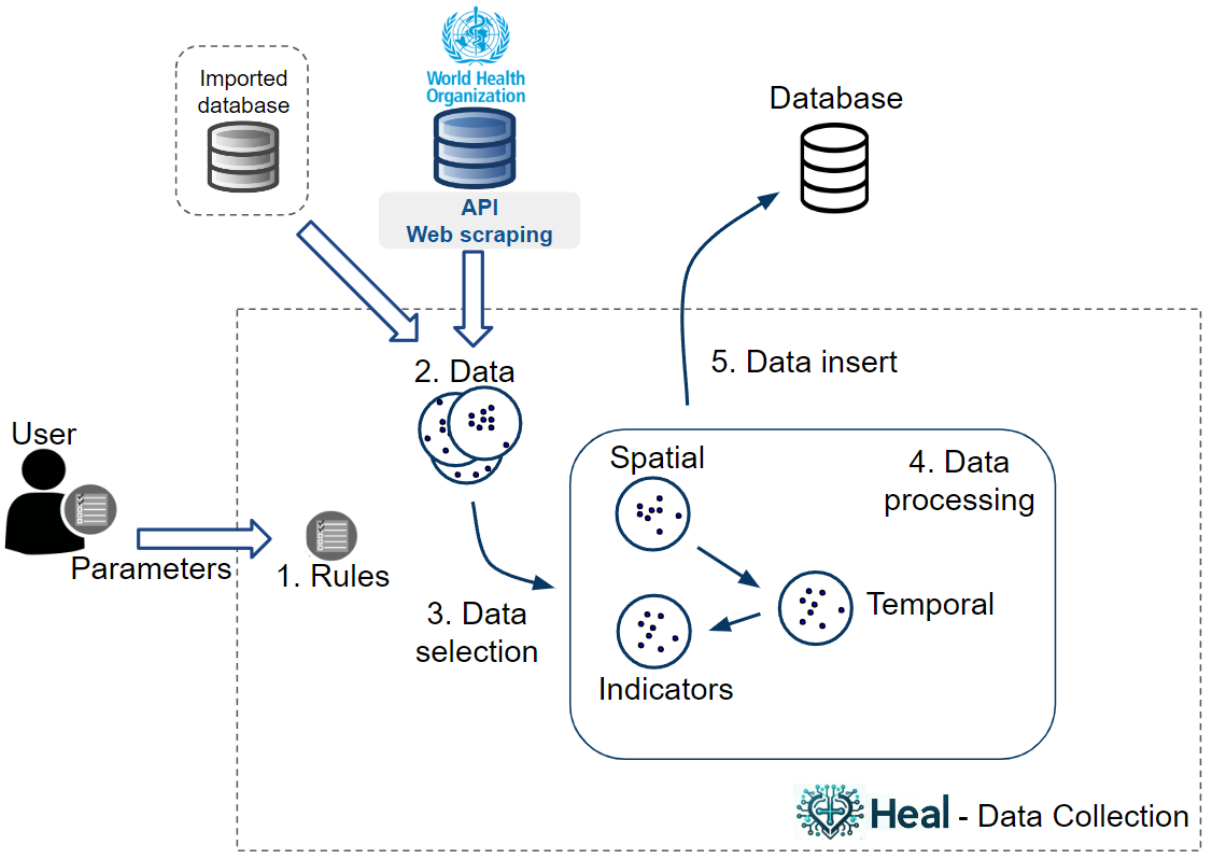
We export the data directly from the WHO website using the API routes provided by the GHO. Then, this data is automated processed to extract the indicators and inserted into the database. Figure 14 illustrates the steps to build the database.

HEAL is enabled to receive parameters as user-defined input about which attributes and dimensions the user would like to import into the system database (step 1 - rules in Figure 14). There is a standard configuration where the user can request the import of all information present in the WHO database. Despite being a step in an AutoML, it is important to provide the user with possibilities to control some minimal tasks, if necessary in specific cases.

After this initial configuration, HEAL can automatically extract all data present in the WHO following user parameters (step 2 - data in Figure 14). After this extraction, the data is processed and information regarding regions, countries, years, indicators and dimensions is organized (step 4 - data processing in Figure 14) and saved in the HEAL database (step 5 - data insert in Figure 14) to be used in the following steps of the pipeline.

In the experiments we carried out, we used the configurations shown in Figure 15, in which we defined the extraction and processing of all WHO indicators and only the

Figure 14 – Steps of Data Collection



Source: Elaborated by the Author.

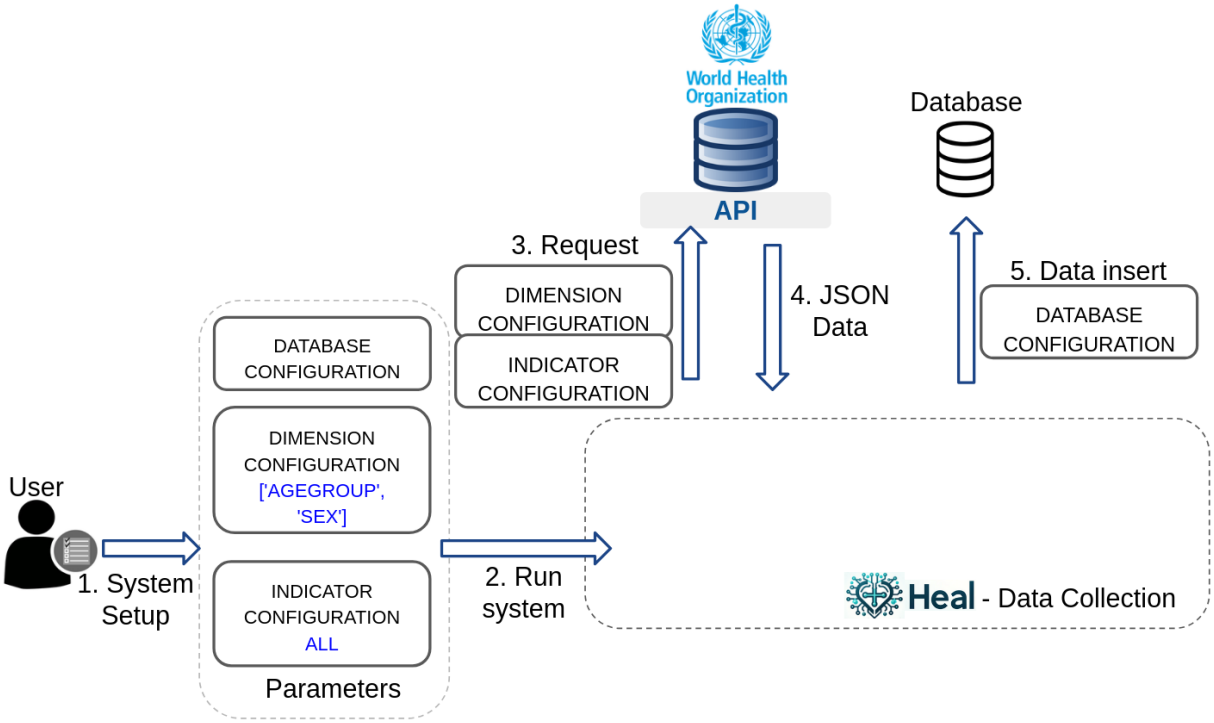
AGEGROUP and *SEX* dimensions (step 1 - system setup and 2 - run system in Figure 15). It's possible to observe that HEAL makes requests to the WHO API based on user-defined parameters and receives JSON containing the data (step 3 - request in Figure 15). From there, it processes this received information to convert it into the required format (represented by the HEAL data collection step in Figure 15, these being the same steps presented in the flow of Figure 14) and saves the information in the database (step 5 - data insert in Figure 15).

The execution of the database construction step required a total of 18,765 seconds (about 5.21 hours) in the test environment used. Thus, we got 2,915 indicators in 90 different years (from 1932 to 2021) and 194 countries for the bases collected and extracted. This processing time, as well as the total indicators, may change according to the environment used and the initial settings made by the user.

In addition, we applied web scraping techniques on the WHO website to enrich our data related to WHO attributes by searching the WHO website[†] for more detailed

[†]https://www.who.int/data/gho/data/indicators/indicators-index. Accessed October 20, 2024

Figure 15 – Configuration in the Data Collection Step



Source: Elaborated by the Author.

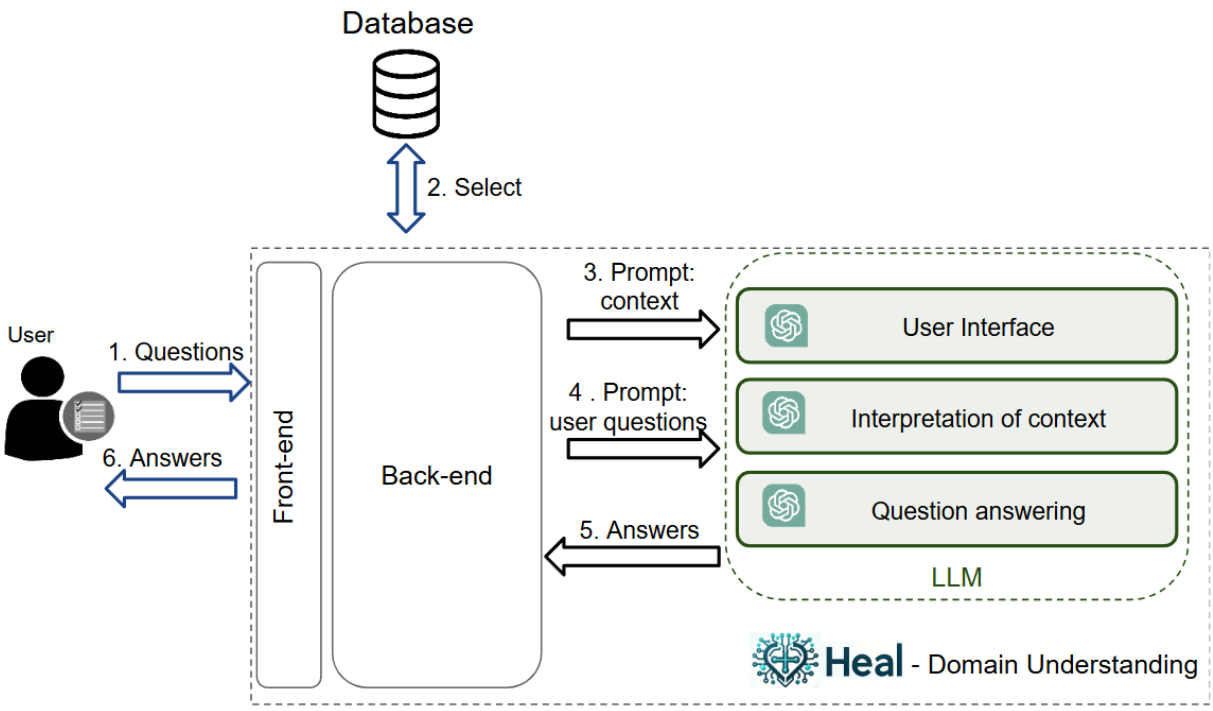
information on each indicator provided, such as: indicator name, short name (code), data type, disaggregation, method of estimation, expected frequency of data, dissemination, definition and rationale.

For web scraping, we used the Beautiful Soup library[‡], searching for the information contained in these fields. This information enriches the HEAL database, so that the LLM application is carried out with more information and, consequently, produces more accurate responses.

After inserting the data extracted from the WHO or receiving the .csv file from the user, we develop a step related to understanding the domain. There are two main objectives in this step, the first being to provide the user with the possibility of interacting with the system in a natural way, thus, the system can interpret the context of the database and the user can learn more about the database and also the possibilities that exist for potential knowledge discoveries. The second objective is to validate the dataset created from the database data and user interactions, ensuring its alignment with the contextual requirements. Figure 16 illustrates the flow responsible for carrying out this step.

[‡]https://pypi.org/project/beautifulsoup4/. Accessed November 05, 2024

Figure 16 – Steps of Domain Understanding



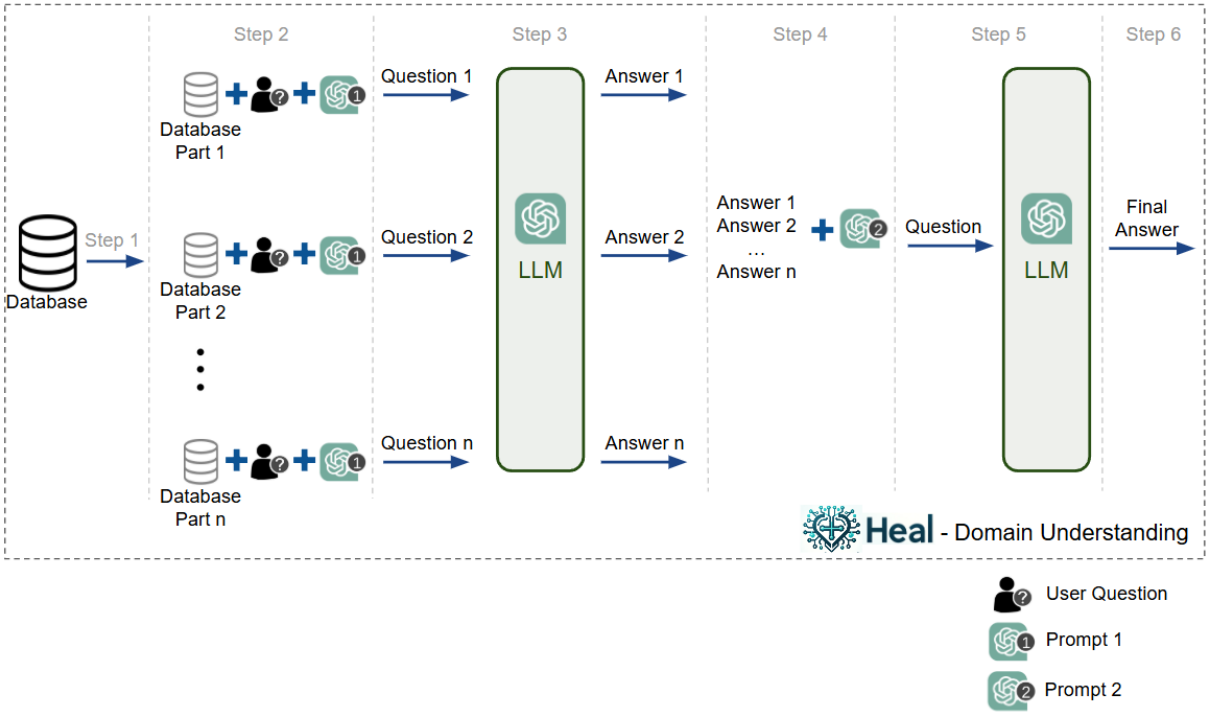
Source: Elaborated by the Author.

The user can ask open-ended questions naturally to the system through a developed front-end (step 1 - questions in Figure 16). When asking questions, HEAL automatically selects information from the database (step 2 - select in Figure 16) and generates prompts corresponding to the questions asked to send them to the ChatGPT-4o API (step 3 - prompt context and 4 - prompt user questions in Figure 16). This LLM will generate responses to the questions (step 5 - answers in Figure 16) and return them for processing and display on the system's front-end (step 6 - answers in Figure 16). The LLM can also validate the dataset extracted from the WHO by checking its integrity and compliance with the user-defined needs and parameters. Furthermore, the questions and observations made by the user can serve as context for the LLM to understand the domain in which the system is operating.

Figure 17 illustrates in detail the steps required to use the LLM ChatGPT-4o API. Therefore, this figure explains steps 3, 4, and 5 of Figure 16.

Firstly, due to the limit of tokens that LLM can process for each request, it is necessary to divide the database into parts, respecting this limit and according to the size of this database (step 1). In step 2, HEAL adds to each of these database parts the user question and prompt 1. The first prompt was created to explain to the LLM its features and provide context about the data sent to them. So, some information was put into this

Figure 17 – Steps to use LLM for Domain Understanding



Source: Elaborated by the Author.

prompt to build the LLM profile:

a) You are a helpful assistant to an AutoML system.

b) In this system, you are helping in the initial stage of domain understanding.

c) You will simulate the behavior of a domain expert in the healthcare area.

d) You will be given some information about the database that the user can explore.

e) Based on this information, you will answer the questions the user asks about that database.

f) Always present the name of the indicators as they are in the original English and always present the indicator code when presenting it to the user.

g) To do this, read the indicators sent previously and work with this data.

h) If you don't know, answer: *I don't know the answer*

i) Always write the answer in English.

Then, this union of the database parts, user question and prompt 1 is sent to the LLM and, for each question sent, the LLM generates a single answer according to each context (step 3). It is then important to merge these responses to create a single answer to the user's question. Therefore, HEAL merges all the answers using a second prompt (step 4) and sends this final question to LLM (step 5), which processes this question and creates a final answer for the user (step 6). Some information is important to build the LLM profile in prompt 2:

a) You are receiving multiple responses that were generated by you when processing different parts of a complete database.

b) Now the goal is to put all these responses together to generate a single answer. Therefore, your answer will be the combination of the previous answers.

c) Interpret the context and see if it will be necessary to add, concatenate or join the responses sent previously.

d) Do not use *I don't know the answer* responses.

e) Always write the answer in English.

f) Format the response into a structure to be presented in html.

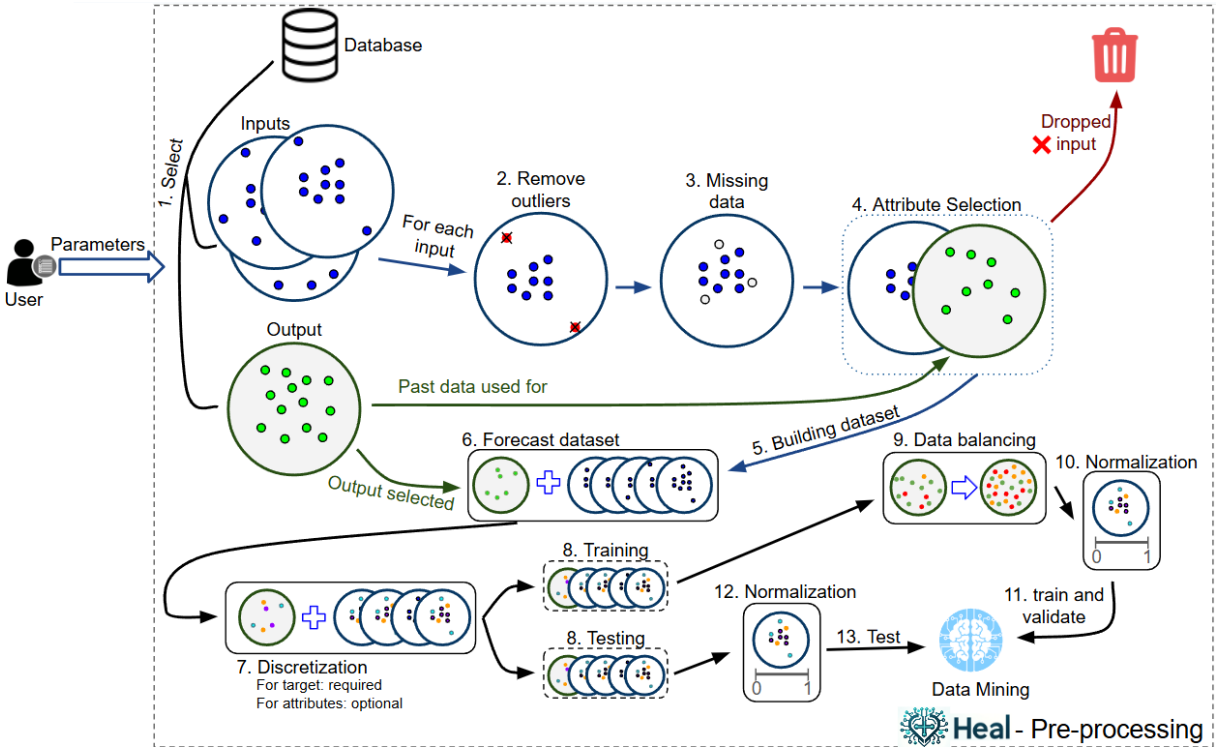
Finally, HEAL allows the user to define a context for knowledge discovery, indicating which attribute is relevant to it along with other settings that the user considers relevant. After this, the system will move on to the next step, responsible for automatically pre-processing the WHO attributes stored in the HEAL database, together with the understanding of the domain acquired by the LLM and the parameterizations made by the user.

5.1.2 Step 2 - Pre-processing

After defining the context based on the parameterization performed by the user, the algorithm applies the steps to define the attributes for building the dataset used for prediction. Therefore, the algorithm analyzes the data for each attribute, checking for missing data, selecting attributes, and removing outliers.

Figure 18 illustrates the steps defined for pre-processing. HEAL is configurable, allowing the user to choose customized options, such as parameters for outliers, missing data analysis, attribute selection parameters for the algorithms, and others. If the user does not define these parameters, HEAL is executed with a default parameterization.

Figure 18 – Steps of Pre-processing



Source: Elaborated by the Author.

After user parameterization, indicators are selected from the database (step 1 - select in Figure 18). Firstly, the outlier identification and exclusion algorithm is applied using the Interquartile Range (IQR) method and the user can choose these parameters in the system (step 2 - remove outliers in Figure 18). For our experiments, we set these values to 0.1 ($Q1$) and 0.9 ($Q3$), respectively.

Attributes with relevant missing data are treated, with two possibilities for this treatment: deletion or imputation (step 3 - missing data in Figure 18). If the definition is by deleting the attribute, HEAL analyzes the attribute dataset and, based on a defined parameter, removes it or not. For example, in the experiments carried out, we parameterized the system to remove attributes with more than 50% missing data. Another option is to perform data imputation, and HEAL is implemented with three algorithms: mean, median and kmeans, and the user can configure by choosing an option to use.

In the next step, the steps for feature selection and feature engineering methods are applied (step 4 - attribute selection in Figure 18). Two different algorithms were implemented for this step: Information Gain and Correlation Matrix. Both can be configured by the user and used to define which attributes will be used in the knowledge discovery process. In the experiments performed to test the system, we set the minimum Information Gain to 2.

In addition, the correlation matrix (Pearson) is also used to compare the previous data of each attribute with the attribute configured as the forecast output. Values close to -1 or 1 indicate a high correlation between the attributes, presenting a linear relationship between them, indicating that the attribute should not be used as input in the prediction algorithm for that proposed output. The cutoff value is also defined by a parameter in the system, and in the experiments we used -0.7 and 0.7.

It is also possible for the user to define which attributes will be analyzed, therefore, the user can define the resources that will be used for the knowledge discovery process. HEAL also addresses redundancies by removing duplicate instances, which may otherwise introduce bias or unnecessarily increase the complexity of the model.

We use empirical testing to define the parameters used in the experiments to determine the efficiency of our system. For each of the tests performed, these steps were followed to define the indicators used for each forecast made. Finally, the system sends the attributes of each dataset resulting (step 6 - forecast dataset in Figure 18) from this pre-processing to the discretization process (step 7 - discretization in Figure 18).

In each forecast that the system performs, the defined output is categorized, dividing the output values by the population of each country, transforming it into a percentage, and later dividing the results into Y different categories (class), which the user can also parameterize.

For discretization, calculations are performed using the lowest value (*Minimum*) and the highest value (*Maximum*) of the characterized data. This *Range* is calculated according to Equation 5.1:

$$Range = (Maximum - Minimum)/Y \tag{5.1}$$

where Y is the number of different classes. Thus, class 1 contains the data between Minimum and Minimum + (*Range**1); class 2 contains the data between Minimum + (*Range**1) and Minimum + (*Range**2); class 3 contains the data between Minimum + (*Range**2) and Minimum + (*Range**3), following up until the final category. This last class contains the data between Minimum + (*Range**(Y-1)) and Maximum. Only the lower bounds are always included, with only the last upper bound also included. Therefore, the problem is converted to a multi-class problem.

Table 8 exemplifies the discretization performed for Y categories. This process characterizes the dataset to divide the data into categories representing all the data. The objective is to send the attributes of each dataset resulting from this pre-processing to the Machine Learning algorithms.

Table 8 – Generalization to create output discretization (using Y categories)

Value Range (button)	Value Range (up)	Class
[Minimum	Minimum + (Range*1))	1
[Minimum + (Range*1)	Minimum + (Range*2))	2
[Minimum + (Range*2)	Minimum + (Range*3))	3
...	...	
[Minimum + (Range*(Y-2))	Minimum + (Range*(Y-1)))	Y-1
[Minimum + (Range*(Y-1))	Maximum]	Y

HEAL makes forecasts of defined categories from the percentage. Thus, in each forecast that the system performs, the defined output can be categorized as described. In the experiments, we used a total of five categories. However, this value can be configured by the system user.

In addition to output discretization, it is possible to apply the discretization process to features according to user parameterization. It is important to highlight that it is also possible to choose the customization of the discretization. Therefore, the method presented in the table presented was not used, and a discretization established by the user was used. Thus, the discretization may be more consistent with real contexts than performing a systematic discretization as presented.

Thus, the dataset is divided into training and testing (step 8 - training and testing in Figure 18). HEAL applies the oversampling or undersampling technique to the training dataset with the aim of artificially increasing the representation of minority classes and, consequently, balancing the defined classes more (step 9 - data balancing in Figure 18. For oversampling, the pipeline uses the Synthetic Minority Oversampling Technique (SMOTE), which creates synthetic examples not only by replicating existing ones, but also by interpolating between similar examples of the minority classes. Oversampling variations that exploit temporal and/or spatial dimensions can also be selected by the user. For undersampling, we use Random Under Sampler, which reduces the number of samples from the majority classes to balance with the number of samples from the minority classes.

Since the data processed by HEAL has a temporal dimension (panel data structure), it is important to offer balancing options that utilize this temporal dimension. Thus, in addition to the traditional methods mentioned, HEAL also offers balancing options based on the spatial and/or temporal dimension (details in Section 2.3), which are:

a) Respecting the temporal dimension (T-SMOTE (ZHAO et al., 2022)): a sliding window is applied over the time series, generating new examples only within consecutive

time windows, which preserves the temporal structure and avoids the creation of unrealistic points that would violate this dependence.

b) Respecting the spatial and temporal dimension: adaptation of the T-SMOTE technique for panel data, also respecting the spatial dimension of the data. Thus, the data is divided among the different spatial features and subsequently T-SMOTE is applied to each spatial dataset.

   For example, if the spatial dimension is related to countries and the temporal region is years, oversampling will be applied to each country separately in a pre-configured time window, such as every 5 years. In this way, the idea of spatial and temporal dimensions is used when applying data balancing.

After balancing, HEAL applies normalization to the training data (step 10 - normalization in Figure 18) and performs the training step on the selected Machine Learning algorithms (step 11 - train and validate in Figure 18). After training and defining the best combination of hyperparameters, the test data is also normalized (step 12 - normalization in Figure 18) and sent to the final testing stage (step 13 - test in Figure 18). Afterward, Machine Learning algorithms are used to predict the defined outputs from the selected indicators and the discretization performed. Therefore, HEAL is intended for classification problems. Despite this, HEAL is fully adjustable and scalable to handle regression issues in the future, with very few changes required.

These steps related to the data mining process are described in the subsequent subsection.

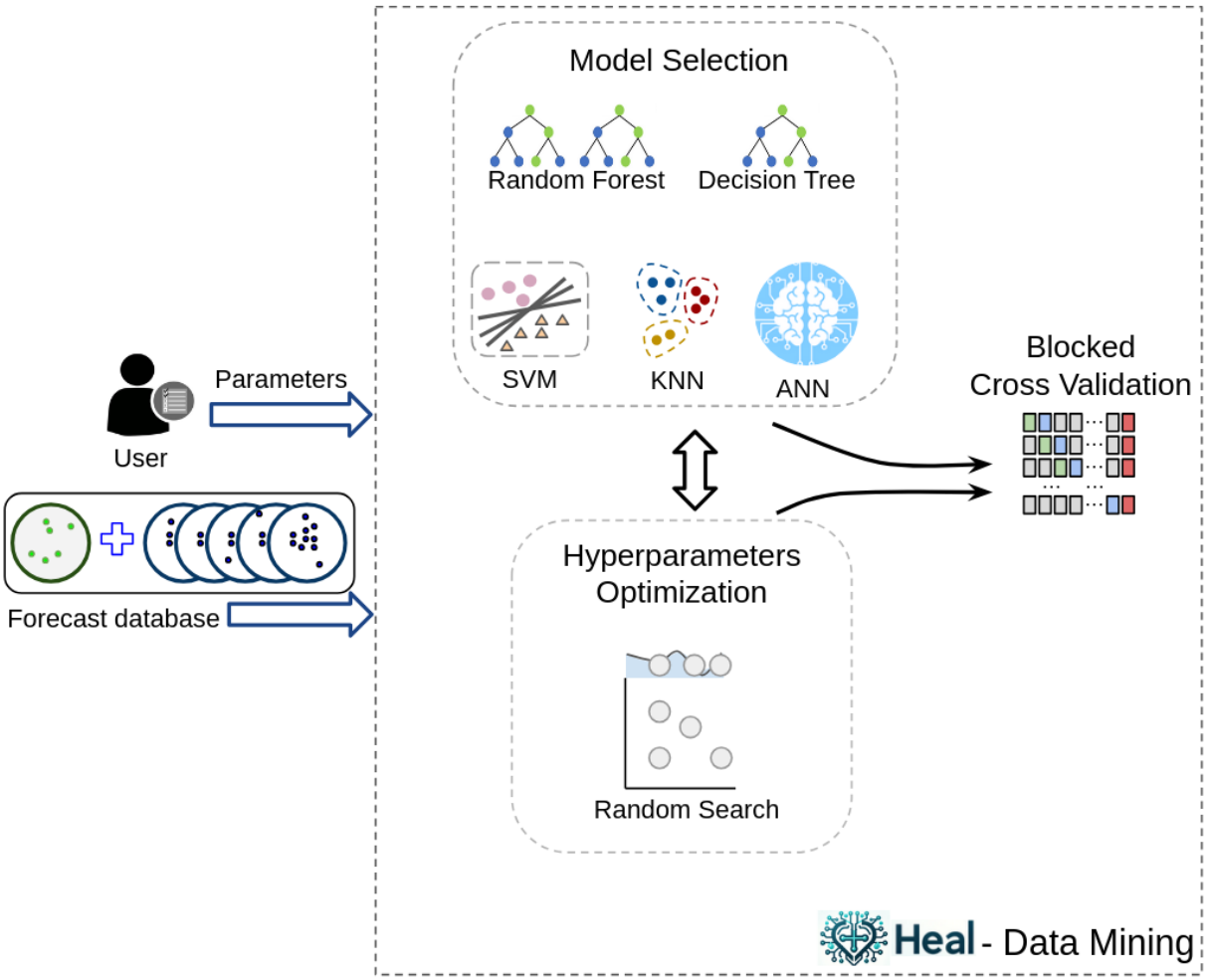
5.1.3 Step 3 - Data Mining

After the data pre-processing steps described in the previous section, this data mining step uses the data generated by the pre-processing and also some possible user-defined parameters, to perform model selection and hyperparameter optimization. Figure 19 illustrates the process carried out in this step.

To this end, Machine Learning algorithms were implemented in HEAL: Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN) and Artificial Neural Network (ANN). As for hyperparameter optimization, we implemented the Random Search method. To validate the results, Blocked Cross Validation was used, respecting the temporal dimension of the data.

Initially, the Combined Algorithm Selection and Hyperparameter Optimization (CASH) method was used, in which several models are created from multiple algorithms and their hyperparameters, seeking to find the one that maximizes the model's perfor-

Figure 19 – Steps of Data Mining



Source: Elaborated by the Author.

mance. To test combinations, we use Random Search, which randomly selects combinations of hyperparameters to test and is able to find good models in a small fraction of the compute time (BERGSTRA; BENGIO, 2012).

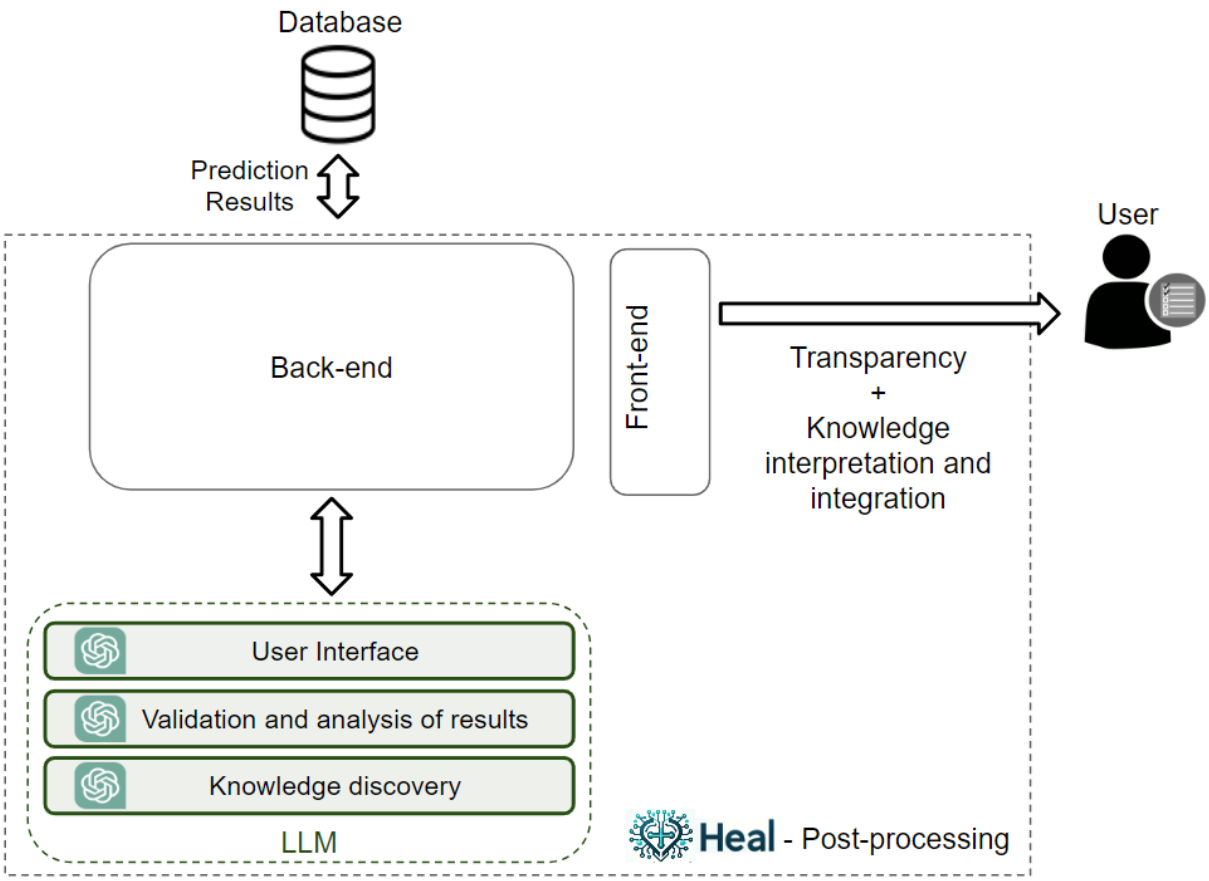
Finally, it is important to highlight that, in order not to execute a black box method, if the user wishes to have more control over HEAL, it is possible for the user to initially define hyperparameter ranges for each algorithm. Furthermore, all results found in each step are saved in a history so that the user can access and understand the process.

The defined combinations are tested using Blocked Cross Validation, which partitions the dataset into test and training/validation and iterates over this second set (always respecting the temporal issue) searching for the best combination of hyperparameters. For each experiment, the system calculates the performance criteria: hits, accuracy, precision, recall, f1-score and the confusion matrix.

5.1.4 Step 4 - Post-processing

Finally, the last step of HEAL aims to provide information about the knowledge discovery process carried out in the previous steps of the HEAL pipeline. Figure 20 illustrates this step.

Figure 20 – Steps of Post-processing



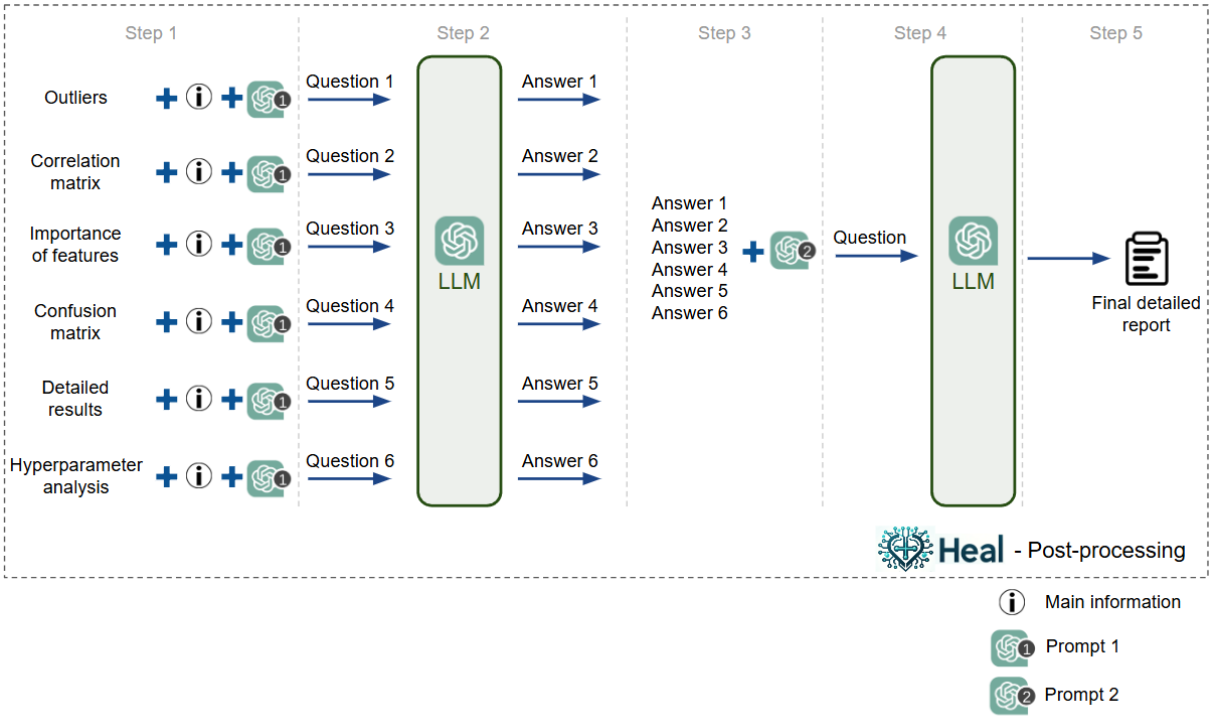
Source: Elaborated by the Author.

For the post-processing step, we developed a dynamic front-end capable of generating a screen with the results and important information for the user.

To complement the final results and assist in the knowledge discovery process, the system sends all found results to ChatGPT-4o following the pattern of a defined prompt. After that, the LLM process provides information and creates a report about the entire executed process. In addition, the LLM processes seeking to discover non-obvious knowledge by analyzing the main attributes defined by each algorithm, the results found and the parameters used.

Figure 21 illustrates in detail the steps required to use LLM in this post-processing.

Figure 21 – Steps to use LLM for Post-processing



Source: Elaborated by the Author.

Firstly, in step 1, HEAL creates different questions to send to the LLM ChatGPT-4o API. These contexts are created using the information generated during the previous steps of the autoML process: outliers, correlation matrix, importance of features, confusion matrix, detailed results and hyperparameter analysis. These contexts are joined with main information (details about the forecasted indicator, such as name, time period, and region) and prompt 1. So, some information was put into this prompt to build the LLM profile:

a) You are a helpful assistant to an AutoML system.

b) In this system, you help with the final post-processing step and then look for useful knowledge discoveries.

c) You will simulate the behavior of a domain expert in the healthcare area.

d) You will receive some information that is the result of a Machine Learning process.

e) Search primarily for non-obvious knowledge discoveries, interpreting the submitted results to assist the user in searching for knowledge discoveries.

f) Always write the answer in English.

In the second stage, each of these questions is sent to the LLM, which provides an answer for each question. After that, to present a detailed final report to the user, it is important to merge all these responses. Therefore, HEAL merges the responses with a second prompt (step 3) to send this final question to LLM (step 4). Some information is important to build the LLM profile in prompt 2:

a) You are receiving multiple responses that were generated from the results found in the AutoML process steps.

b) Now the goal is to put all these responses together to generate a single answer. Therefore, your answer will be the combination of the previous answers.

c) Interpret the context and see if it will be necessary to add, concatenate or join the responses sent previously.

d Write the results found in report format.

e) Always write the answer in English.

f) Write the response in an HTML text format to be used within a website that is already HTML compliant.

Finally, the LLM provides a final response in the form of a detailed final report (step 5). This final report is presented in our front-end to help the user make decisions and discover knowledge.

Finally, this interface presents the user with all the information and data relating to each of the steps taken (each of the responses generated separately), presenting a complete record of each step, in addition to presenting the final detailed report. Therefore, the proposal is that HEAL is not a black box method, being different from most works found in the literature. HEAL presents all the information from all the steps, giving the user more control and making it easier to understand what was done throughout the process. This can also help the user identify errors and issues during each step, which can help correct them in future runs.